

Academic Publication Dossier & Consolidated Bibliography

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1 Executive Research Overview

This dossier centralizes the systems engineering and formal verification research developed by Xavier Callens across public and private codebases. Operating at the intersection of bare-metal operating system safety, formal computer-checked mathematics (Lean 4), high-performance solvers, and accelerated machine learning runtimes (RISC-V & Google TPUs), this body of work addresses the fundamental “safety-versus-performance” tax.

By utilizing Rust’s affine type system for zero-cost resource management, and bounding hardware-abstraction boundaries via automated theorem proving, these projects prove that mathematical security guarantees do not concession, but rather unlock, compiler-guided systems optimizations.

2 Directory of Research Repositories

All compiled systems software, mathematical proofs, and automated benchmarks are tracked across the following repositories:

Repository Name	Type	Core Focus / Subsystems
rust-linux-mini-kernel	Private	MVK Bare-metal Kernel, Security Proofs, RunuX EuroSys Stack
runux-ai-runtime	Private	Unified HAL, PolarQuant, Systolic StableHLO compiler kernels
x-rusty-SUNDIALS	Private/Public	rusty-SUNDIALS Workspace, CVODE Solver, Newton Verification
xthesis	Private	Academic Thesis, Neuro-Symbolic Orchestrator, EDDBA Vignettes

Table 1: Xavier Callens’ Research and Code Repositories.

3 Consolidated Publications & Articles

3.1 1. RunuX-AI: Systolic-Aware ML Runtimes

Status (2026-09-26): the full paper is quarantined at `paper/quarantine/runux-ai.paper.tex` pending confirmation that `scripts/run_benchmarks.sh --simulate-tpu` reproduces the figures below; see `docs/roadmap/PAPER_VERIFICATION_TODO.md`.

Title: RunuX-AI: A Unified, Memory-Safe Rust Runtime for High-Efficiency Federated LLM Execution across Edge RISC-V and Cloud Systolic TPUs

Target Venue: MLSys ’27 / IEEE Transactions on Parallel and Distributed Systems (TPDS)

Repository Reference: Private, github.com/xaviercallens/runux-ai-runtime

Core Contributions:

- **Unified Memory-Safe HAL:** Trait-based, compile-time backend generic dispatch spanning RISC-V Vector processors and Cloud TPU PJRT.

- **PolarQuant & QJL:** Lossy 3-bit block-wise KV-cache quantization using orthogonal Householder rotations and scale error correction, achieving < 0.05 reconstruction error (simulated).
- **Systolic-Aware StableHLO Tiling:** Programmatic StableHLO matrix multiplier loops matching TPU v5e's 128×128 MXU boundaries, securing an **88.0% occupancy rate** (173.4 TFLOPS) **under simulation, not measured on physical TPU v5e hardware.**
- **Green AI Speculative Scheduler:** A power-aware sampling scheduler that dynamically balances speculation budgets (K) against regional grid carbon intensities (France, USA, China).

3.2 2. RunuX: Linux Kernel Translation

Title: RunuX: A Formally Verified, FFI-Compatible Rust Translation of the Linux Kernel Networking Stack

Target Venue: ACM EuroSys '27 (Accepted/Under Review)

Repository Reference: Private, github.com/xaviercallens/rust-linux-mini-kernel/tree/main/paper

Core Contributions:

- **297 Translated Modules:** First comprehensive, FFI-compatible Rust translation of the core Linux 5.10 LTS networking stack using binary-compatible `#[repr(C)]` FFI bindings.
- **High-Fidelity Chaos Testing:** Statistically identical native boot and packet latency, with **zero panics/oops** under extensive Chaos Mesh injection (partitions, losses, CPU stalls) on GKE.
- **Lean 4 Formal Verification:** 432 machine-checked theorems across 36 specification modules (138 axioms, 245 open proof obligations as of 2026-09-26); a dedicated 84-theorem Ring-0 isolation model is fully closed with zero `sorry` and zero axioms. See `paper/runux_paper.tex` §4 for the measured breakdown and the oracle-gated proof-completion methodology used to reduce the open count.

3.3 3. MVK: The Bare-Metal Architecture

Status (2026-09-26): the full paper is quarantined at `paper/quarantine/mvk_scientific_paper.tex` – no reproducible `rdtsc` benchmark harness for the figures below was found in this repository; see `docs/roadmap/PAPER_VERIFICATION_TODO.md`.

Title: Beyond Legacy Abstractions: A Formally Verified, High-Performance Bare-Metal Rust Architecture (MVK)

Target Venue: ACM SOSP / ACM Transactions on Computer Systems (TOCS)

Repository Reference: Private, github.com/xaviercallens/rust-linux-mini-kernel

Core Contributions:

- **The MVK Kernel:** A 64-bit bare-metal operating system architecture in Rust featuring lock-free Atomic Reference Counting and Sv39/Sv48 paging.
- **Aggressive Optimizations:** Leverages lack of pointer aliasing in safe Rust to allow the LLVM backend to unroll loops, reducing `mmap` latency by **25.3%** and page fault overhead by **12.2%**.
- **Zero Latency Drift:** Demonstrates complete immunity to SLAB fragmentation, yielding flatline $45\mu\text{s}$ execution loops under 2-hour sustained 100% CPU stress tests.

3.4 4. MVK: Cyber-Security & Exploit Eradication

*Status (2026-09-26): the full paper is quarantined at `paper/quarantine/security_scientific_paper.tex` – the “100% elimination” claim is contradicted by a live aliasing bug found this session in `crates/ai_detector` and by 722 `unsafe` blocks workspace-wide with only $\sim 14\%$ **SAFETY**: comment coverage; see `docs/roadmap/PAPER_VERIFICATION`*

Title: The Death of the Kernel Zero-Day: Mathematically Eliminating Spatial and Temporal Memory Vulnerabilities in Bare-Metal Architectures

Target Venue: IEEE Symposium on Security and Privacy (S&P) / Dependable Systems

Repository Reference: Private, github.com/xaviercallens/rust-linux-mini-kernel

Core Contributions:

- **Spatial and Temporal Immunity:** Structural proof that Rust’s lifetime borrow checker eradicates Use-After-Free conditions, and slice indexing validates bounds at compile-time.
- **RCU Data Race Prevention:** Prevents TOCTOU concurrency exploits using a custom compile-time verified lock-free RCU pointer primitive.
- **95% Attack Surface Reduction:** Quarantines raw hardware IO to $< 4.2\%$ of the codebase, leaving the rest verified as purely safe.

3.5 5. rusty-SUNDIALS: High-Performance Solver

Title: rusty-SUNDIALS: A Memory-Safe Rust Implementation of CVODE with Formally Verified Newton Convergence

Target Venue: ACM Transactions on Mathematical Software (TOMS)

Repository Reference: Private/Public, github.com/xaviercallens/x-rusty-SUNDIALS

Core Contributions:

- **Rust CVODE Solver:** complete BDF 1–5 and Adams-Moulton 1–12 implementations with builder-pattern configuration.
- **Newton Convergence Bug Fix:** Discovery and fix of LLNL reference’s convergence rate persistence bug, reducing iteration counts per step from 2.42 to **1.40** (exceeding C reference’s 1.44).
- **Refinement Proofs:** Formalization of $C \leftrightarrow \text{Rust}$ behavioral equivalence via Lean 4 proof terms covering the entire nonlinear solver surface.

4 BibTeX Database Reference

For citing these publications in dissertation manuscripts or related literature reviews, utilize the following standard BibTeX entries:

```
1 @inproceedings{CallensMvk2026,
2   author      = {Callens, Xavier},
3   title       = {Beyond Legacy Abstractions: A Formally Verified, High-
4     Performance Bare-Metal Rust Architecture (MVK)},
5   booktitle   = {Proceedings of the ACM Symposium on Operating Systems
6     Principles (SOSP)},
7   year        = {2026},
8   note        = {MVK Bare-metal OS kernel in safe Rust verified via Lean
9     4.},
10  url          = {https://github.com/xaviercallens/rust-linux-mini-kernel}
11 }
12
13 @inproceedings{CallensSecurity2026,
```

```

11 author      = {Callens, Xavier},
12 title       = {The Death of the Kernel Zero-Day: Mathematically
    Eliminating Spatial and Temporal Memory Vulnerabilities in Bare-
    Metal Architectures},
13 booktitle   = {IEEE Symposium on Security and Privacy (S\&P)},
14 year        = {2026},
15 note        = {Formal spatial/temporal safety proofs in Lean 4. Attack
    surface reduction >95\% (unsafe <4.2\%).},
16 url         = {https://github.com/xaviercallens/rust-linux-mini-kernel}
17 }
18
19 @inproceedings{CallensRunux2027,
20   author      = {Callens, Xavier},
21   title       = {RunuX: A Formally Verified, FFI-Compatible Rust
    Translation of the Linux Kernel Networking Stack},
22   booktitle   = {Proceedings of the ACM European Conference on Computer
    Systems (EuroSys)},
23   year        = {2027},
24   note        = {297 translated modules, validated via QEMU, libFuzzer,
    and GKE Chaos Mesh stress tests.},
25   url         = {https://github.com/xaviercallens/rust-linux-mini-kernel}
26 }
27
28 @article{CallensSundials2026,
29   author      = {Callens, Xavier},
30   title       = {rusty-SUNDIALS: A Memory-Safe Rust Implementation of
    CVODE with Formally Verified Newton Convergence},
31   journal     = {ACM Transactions on Mathematical Software (TOMS)},
32   year        = {2026},
33   note        = {TOMS draft. Resolves LLNL convergence rate persistence
    bug. Verified via Lean 4.},
34   url         = {https://github.com/xaviercallens/x-rusty-SUNDIALS}
35 }
36
37 @inproceedings{CallensRunuxAi2027,
38   author      = {Callens, Xavier},
39   title       = {RunuX-AI: A Unified, Memory-Safe Rust Runtime for High-
    Efficiency Federated LLM Execution across Edge RISC-V and Cloud
    Systolic TPUs},
40   booktitle   = {Proceedings of the Conference on Machine Learning and
    Systems (MLSys)},
41   year        = {2027},
42   note        = {MLSys '27 draft. Trait-based HAL, PolarQuant (3-bit
    rotations + QJL), Systolic tiling, Green AI Speculative Scheduler.},
43   url         = {https://github.com/xaviercallens/runux-ai-runtime}
44 }

```

Listing 1: BibTeX database entries for Xavier Callens’ articles.